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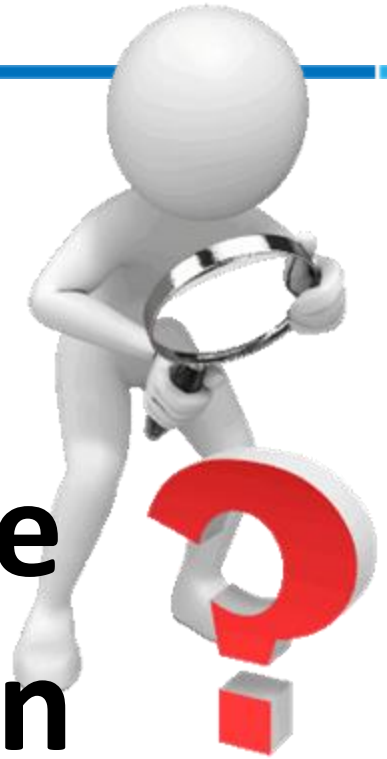
UNIVERSIDAD YACHAY TECH

Science Communication

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Question



What is science communication

Science Communication



It is a set of activities associated with the production and use of information, from the moment the scientist conceives an idea until the information on the research results is published.

Science Communication

The main objective of scientific communication is the integral management of knowledge including:

- Knowledge Registration
- Rigorous Evaluation
- Dissemination
- Knowledge Accumulation

Focus on Facts and Human Perceptions

Science Communication: general context

The Declaration on Science and the Use of Scientific Knowledge adopted in the UNESCO sponsored World Conference on Science in 1999, established the following:



1. Scientific knowledge should be shared.

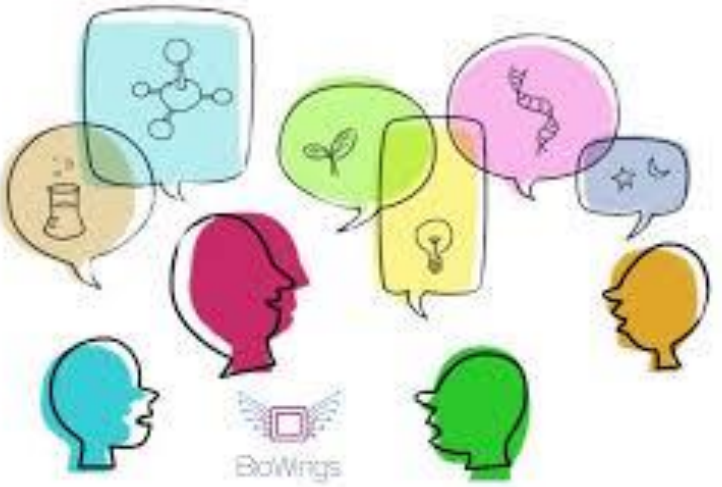
2. There is need for real cooperation between governments, civil society, the business sector and scientists.

3. Scientists should be governed by ethical standards.

4. Professionals who circulate scientific information see communicating with the public as a vital responsibility.

Dissemination and Disclosure

The dissemination of scientific knowledge allows research results to be known, discussed and shared with the academic community.





Disclosure refers to the act of making information known or public, especially information that was previously private or unknown.

Scientific communication: dissemination

Features

Provide the scientist with the best answers to specific questions.

Contribute so that the scientist knows about the new developments in his field of action.

Encourage the scientist to take new knowledge, in addition to that of his area of interest.

Disseminate the main trends in emerging areas, providing scientists with an idea of the relevance of their work.

Scientific communication: dissemination

Features

Test the reliability of the new knowledge, before the possibility of testimonies and verifications.

Redirect or expand the role and interest of scientists.

Provide feedback for the improvement of the researcher's production.

Science communication: disclosure

Features

The production of knowledge occurs from previous results.

Researchers consume and communicate information from the beginning to the end of their studies.

Science depends on the wide dissemination of what is produced.

Science depends on communication processes.

Science communication: disclosure

Feature	
	Promote curiosity towards a subject or field of knowledge.
	Offer the information necessary for society to acquire an opinion on issues related to advances in science and their repercussions.
	Help understand the social, political or cultural transformations of new events and facilitate each person to have critical thinking.

What is scientific disclosure for?



Reduces the educational gap in the different population sectors

It works as an educational element that facilitates the population's access to academic, technical or specialized information.

People can form an opinion on a controversial or difficult issue, and be able to act on it.

La Presentación de tu Vida (29, septiembre, 2019). Cómo hacer DIVULGACIÓN CIENTÍFICA [Archivo de video]. Recuperado de: https://www.youtube.com/watch?v=UWDsWwG0BAI

Dissemination versus scientific disclosure

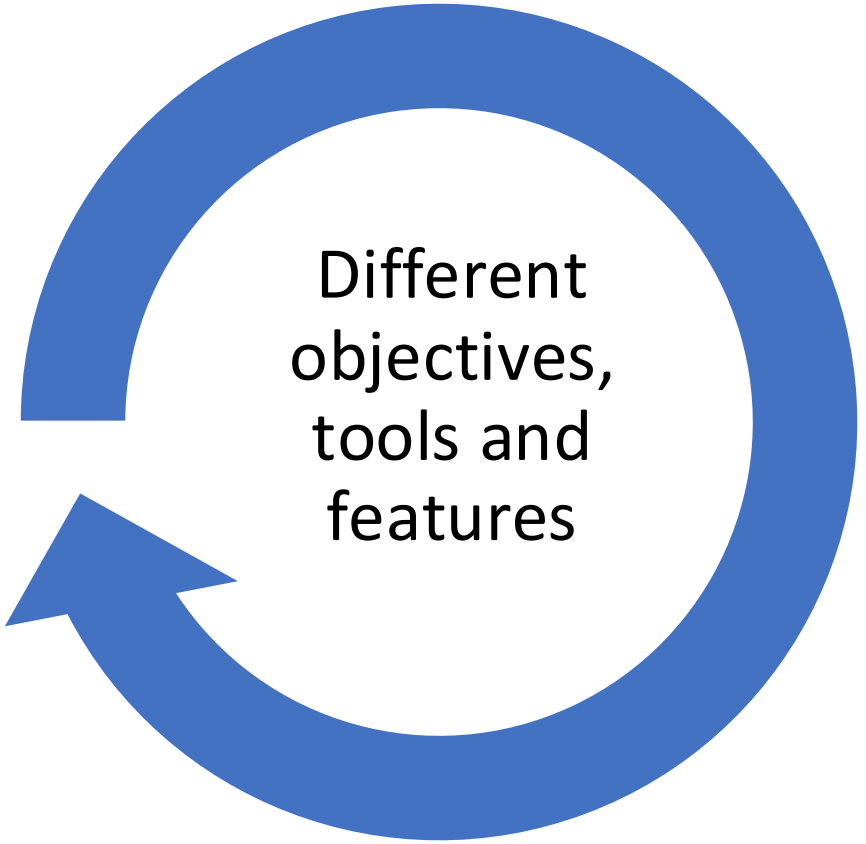
Dissemination	Disclosure
Communication between "peers", that is, between specialists.	Communication between "non-peers", partners do not share the same knowledge.
It establishes a symmetrical relationship between the subject who communicates and the individual who interprets.	Interdisciplinary work that communicates knowledge to various audiences.
It proposes a delimited and specific socio-professional or technical framework.	General audience.

Who should say science?

There is a consensus on the need to **bring scientific knowledge to society** and get educated and informed citizens, who support science and its financing and have sufficient criteria and opinion on issues that affect everyone.



- **Scientists**
  Journalists
  Disclosers
  Institutions / companies



Scientists

Credibility | Knowledge | Authority



They assume an essential role in the dissemination of science, since they are the top experts in their specialty.

- They have found outreach to be an interesting and fun activity, they consider it part of their job and do it on a regular basis.
- They use new technologies, they communicate quickly, directly and on a daily basis.
- Dissemination channels: blogs, social networks, articles, conferences and books.

Journalists

- Diffusion channels: traditional media (TV, press and radio), digital media, blogs, podcast and social networks.
- Professionals with the ability to tell attractive stories that demonstrate to citizens the consequences of scientific knowledge in their lives.
- It is responsible for translating knowledge so that people can learn clearly.



Historically they are the actors in the dissemination of science.

Scientific disseminator

Scientific dissemination can be carried out by **non-scientific disseminators and scientific disseminators.**

In either case, publishers must maintain these principles in their disclosure product:

- Be faithful to the scientific content.
- Be aimed at a **non-specialized audience.**
- Create a new message with **non-technical and contextualized language** to make it accessible.
- Express oneself under the agreement of a **voluntary public.**

Scientific disseminator



B A L A N C E

The essential tension of science popularization is based on these principles:

A) A product close to science requires technical language, therefore it needs more context.

B) A product with pleasant language has a probability of errors.

Companies / institutions

Institutions related to science.



Companies in the technological or innovative field.



- They are important actors in the popularization of science.
- Its mission is to provide any interested party with immediate and easy access to information and documentation in their area.
- They use the web pages, open houses and other activities.

What does the media work for?



Human beings need the media to understand the world around them.

They are social animals and are increasingly interested in knowing what happens beyond their professional and family life.

They seek to know what happens in the political, economic, cultural, sports, etc.

Out of curiosity, but also because these areas directly or indirectly influence our lives.

Communication theories: Agenda setting

- **McCombs et al. (2014)**

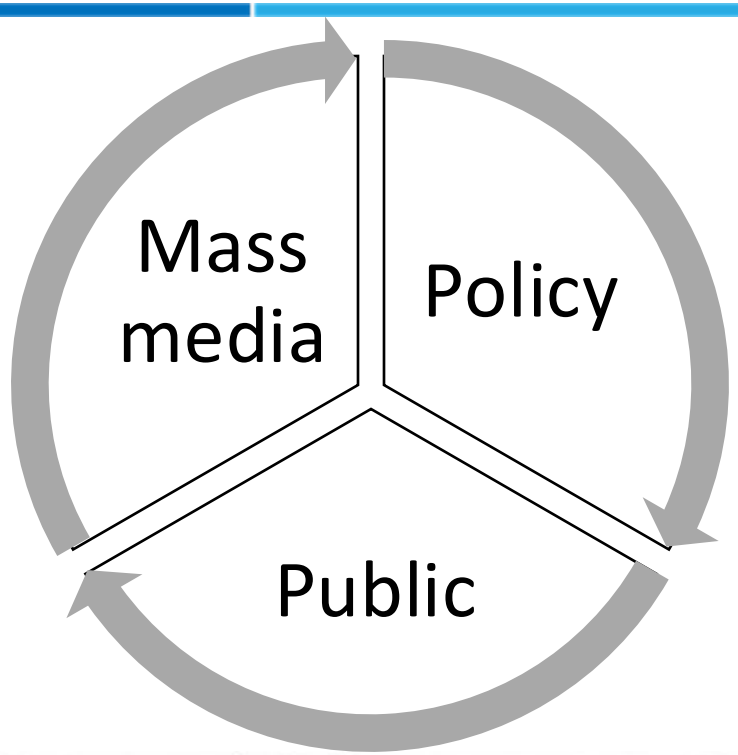
The works carried out on the Theory of the agenda-setting from different aspects reach the same outlet, in which the media have the ability to create reality.

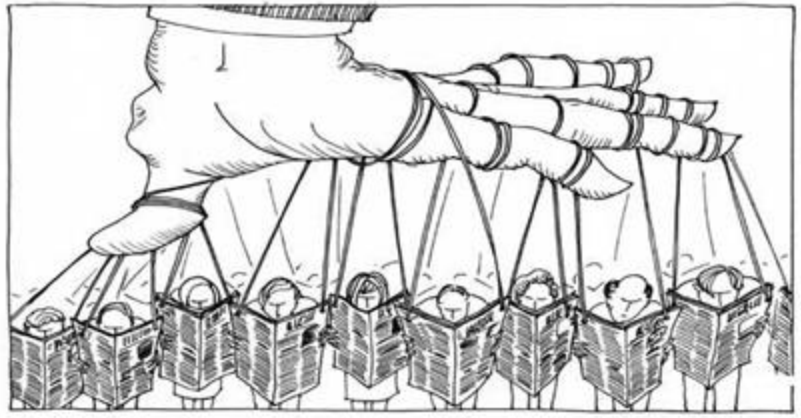
- **Santillán (2018)**

They claim that the formation of public opinion is a consequence of the role of the media.

- **Tirado-Espín, A., et al. (2020).**

The determination of a set of news makes citizens perceive that reality as the most important of its day.





Communication theories: Framing

- **McCombs y Shaw (1972)**

The news not only marks the relevance or importance of the social issues proposed by the agenda. That is why this theory is known as the second level of agenda setting theory.

- **Entman (1993)**

People's way of thinking about certain issues is manipulated, depending on how you put them. This way of presenting information is known as a news frame or frame.

- **Aday (2006) y Keum et al. (2005)**

These frames have effects on the cognitive responses of the subjects as well as on people's attitudes and beliefs about the issues discussed.



Media features

Report

The events that occur are told.

NEW

- Investigation process.
- Contrasting of sources.
- Deontological codes.
- Audiovisual and written media.



Media features

Explain

What happens in the world is necessary to explain the context in which the events occur.

REPORTAGE



Vision as wide and varied as possible of the subject in question.

The space / time available is usually much greater.

The journalist must refrain in these genres from including their own opinions directly.

Funciones de los medios de comunicación

Opine

People need to be told what is happening in the world so that they can build an opinion of that reality.

The media make the task easier by including different types of text or genres.

EDITORIAL

ARTICLE

COLUMN



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¡Thank you!

ESCANÉAME

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